

Wenbo Lyu

B.Sc. Graduate

Curriculum Vitae

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Education and qualifications

B.Sc. In Geographic Information Science Shaanxi Normal University 2025

Research Experience

2026.6- Research Assistant, The Hong Kong Polytechnic University
2025.9-2026.6 Research Assistant, Eastern Institute of Technology, Ningbo
2024.8-2025.8 Research Intern, The Hong Kong University of Science and Technology (Guangzhou)

Research interests

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.

Selected research papers

(Lyu, Dai, et al. 2026); (Lyu, Lei, et al. 2026); (Lv et al. 2025); (Lv et al. 2024); (Li et al. 2026); (Xu et al. 2026); (Chen et al. 2025)

Note: Publications under "Lyu, W." and "Lv, W." both refer to Wenbo Lyu.

Chen, Chunjiang, Yongze Song, Wenbo Lv, et al. 2025. "Predicting Pavement Cracking Performance Using Laser Scanning and Geocomplexity-enhanced Machine Learning." *Computer-Aided Civil and Infrastructure Engineering*, ahead of print, April. <https://doi.org/10.1111/mice.13489>.

Li, Bingchen, Wenbo Lv, Hu Ding, Yongze Song, and Wufan Zhao. 2026. "New-Type Urbanization Policy, Urban Expansion, and Employment Quality in Chinese Cities: Evidence from a Quasi-Natural Experiment in China." *GIScience & Remote Sensing* 63 (1): 2716413. <https://doi.org/10.1080/15481603.2026.2716413>.

Lv, Wenbo, Yangyang Lei, Fangmei Liu, Jianwu Yan, Yongze Song, and Wufan Zhao. 2025. "gdverse: An R Package for Spatial Stratified Heterogeneity Family." *Transactions in GIS* 29 (2): e70032. <https://doi.org/10.1111/tgis.70032>.

Lv, WenBo, FangMei Liu, Kai Cai, et al. 2024. "Distinguishing the Impacts and Gradient Effects of Climate Change and Human Activities on Vegetation Cover in the Weihe River Basin, China." *Journal of Geophysical Research: Biogeosciences* 129 (10): e2024JG008297. <https://doi.org/10.1029/2024jg008297>.

Lyu, Wenbo, Shaoqing Dai, Yongze Song, et al. 2026. "Measuring Causal Strengths from Spatial Cross-Sectional Data with Geographical Cross Mapping Cardinality." *International Journal of Geographical Information Science*, 1–23. <https://doi.org/10.1080/13658816.2026.2687121>.

Lyu, Wenbo, Yangyang Lei, Wen Yi, et al. 2026. "Causal Discovery in Urban Data with Temporal Empirical Dynamic Modeling: The R Package tEDM." *Computers, Environment and Urban Systems* 127 (July): 102435. <https://doi.org/10.1016/j.compenvurbsys.2026.102435>.

Xu, Lingzi, Haochen Shi, Wenbo Lyu, Yan Zhang, and Hongbo Zhao. 2026. "Beyond Correlation: Unveiling the Causal Link Between Street-Scapes and Recreational Walking/Cycling." *Journal of Transport Geography* 136: 104789. <https://doi.org/10.1016/j.jtrangeo.2026.104789>.